

QAOA does not update the operator when calling `construct_circuit`

<https://github.com/Qiskit/qiskit/issues/7223>

ROW 90

QAOA does not update the operator when calling construct_circuit #7223

New issue

Closed

#7225



EliasCombarro opened on Nov 3, 2021

...

Information

- **Qiskit Optimization version:** The one on the IBM Quantum Experience
- **Python version:** The one on the IBM Quantum Experience
- **Operating system:** The one on the IBM Quantum Experience

What is the current behavior?

If you call `construct_circuit` with two different operators, the second time the operator is not updated and the operator of the first call is used instead.

Steps to reproduce the problem

```
from qiskit.algorithms import QAOA
from qiskit.opflow.primitive_ops import PauliOp
from qiskit.quantum_info import Pauli
from qiskit import Aer
```

```
op = PauliOp(Pauli("ZZ"))
qaoa = QAOA()
qaoa.construct_circuit([0,0],op)[0].draw(output="text")
```

```
op = PauliOp(Pauli("IZ"))
qaoa.construct_circuit([0,0],op)[0].draw(output="text")
```

What is the expected behavior?

The second call to `construct_circuit` should update the operator and return the correct circuit

Suggested solutions



Assignees

No one assigned

Labels

bug mod: algorithms

Type

No type

Projects

No projects

Milestone

0.19
Closed on Dec 6, 2021, 100% complete

Relationships

None yet

Development

Fix QAOA handling of ``PauliSumOp`` identities and changing cost operators
Qiskit/qiskit

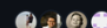
Notifications

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Participants



t-imamichi on Nov 3, 2021

Member ...

Thank you for your report. Because `QAOA` is part of Terra, could you report it there? <https://github.com/Qiskit/qiskit-terra/issues>



woodsp-ibm transferred this issue from [qiskit-community/qiskit-optimization](#) on Nov 3, 2021



woodsp-ibm added `mod: algorithms` `bug` on Nov 3, 2021



woodsp-ibm on Nov 3, 2021

Member ...

It seems this comes about since QAOA extends VQE and while there is some additional check in there for QAOA the anstaz is not built out again if the number of qubits is the same when the operator changes. Eg. change your second call above to

```
op = PauliOp(Pauli("IIZ"))
qaoa.construct_circuit([0,0],op)[0].draw(output="text")
```



and it works as expected.



 **woodsp-ibm** transferred this issue from [qiskit-community/qiskit-optimization](#) on Nov 3, 2021

 **woodsp-ibm** added **mod: algorithms** **bug** on Nov 3, 2021



woodsp-ibm on Nov 3, 2021

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t-imamichi on Nov 4, 2021 · edited by t-imamichi

Edits ... Member ...

I see. We should compare `operator` and `cost_operator` in the following line. I will make a PR to fix it.

https://github.com/Qiskit/qiskit-terra/blob/0a4c666ca09e129768f83f7c95ee7168455d869d/qiskit/algorithms/minimum_eigen_solvers/qaoa.py#L130



 **t-imamichi** mentioned this on Nov 4, 2021

 [Fix QAOA handling of PauliSumOp identities and changing cost operators #7225](#)

 **Cryoris** closed this as **completed** in [#7225](#) on Nov 4, 2021



EliasCombarro on Nov 4, 2021

Author ...

Thank you



1

 **kdk** added this to the [0.19](#) milestone on Nov 15, 2021

Fix QAOA handling of PauliSumOp identities and changing cost operators #7225

<> Code

Merged

Cryoris merged 1 commit into Qiskit:main from t-imamichi:fix-qaoa on Nov 4, 2021

Conversation 2

Commits 1

Checks 0

Files changed 5

+41 -4

t-imamichi commented on Nov 4, 2021 · edited

Member

Summary

Fixes following two bugs.

- QAOAAnsatz cannot handle identities of PauliSumOp and it has a wrong number of parameters.
- QAOA.construct_circuit does not update circuits if different operators with the same number of qubits are given.

Fixes #7223

Details and comments

I introduced QAOA._cost_operator to check whether operator is same as _cost_operator or not. If we don't have to decompose QAOA.ansatz, we can compare operator with QAOA.ansatz.cost_operator. Once #6674 is fixed, we need to revisit it.
https://github.com/Qiskit/qiskit-terra/blob/0a4c666ca09e129768f837c95ee7168455d869d/qiskit/algorithms/minimum_eigen_solvers/qaqa.py#L133

Fix QAOA bugs

d4a5aeb

t-imamichi requested review from manaelmarques, woodsp-lbm and a team as code owners 4 years ago

coveralls commented on Nov 4, 2021

Pull Request Test Coverage Report for Build 1420730151

- 6 of 6 (100.0%) changed or added relevant lines in 2 files are covered.
- 3 unchanged lines in 1 file lost coverage.
- Overall coverage increased (+0.001%) to 82.376%

Files with Coverage Reduction	New Missed Lines	%
qiskit/pulse/library/waveform.py	3	90.38%

Totals

coverage 82%

Change from base Build 1419013932: 0.001%

Covered Lines: 49662

Relevant Lines: 60287

Coveralls

Reviewers

Cryoris

manaelmarques

woodsp-lbm

Assignees

No one assigned

Labels

Changelog: Bugfix mod: algorithms

Projects

None yet

Milestone

0.19

Development

Successfully merging this pull request may close these issues.

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4 participants